Veronica Ramos Rodriguez

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Education

Vanderbilt University — Nashville, TN
PhD in Electrical and Computer Engineering — *Aug. 2026 – May 2031*
NSF GRFP & GEM Employer Fellow

University of Florida — Gainesville, FL
Bachelor's in Biomedical Engineering — *Jun. 2022 – May 2026*
Minors in Electrical Engineering and Digital Arts & Sciences
Technical Focus in Computer Science

Relevant Coursework
VLSI Circuit Design, Stochastic Processes, Data Structures & Algorithms, Machine Learning, Bioelectric Systems, Biosignals & Systems, Cyber-Physical Security, Biomedical Image Processing, Medical Imaging Technology, Game Engine Development

Experience

VLSI Design Teaching Assistant — Aug. 2026 – Present
Vanderbilt University — *Nashville, TN*
- Assist 35 students with Cadence Virtuoso layouts, schematics, and simulations of gates and computer memory.

Edge Computing Systems Intern, GEM Fellow — May 2026 – Aug. 2026
SLAC National Accelerator Laboratory — *Menlo Park, CA*
- Developed an autonomous experiment steering system for serial femtosecond crystallography research using a local LLM built on a RAG database of 5,000+ files to provide geometry corrections, operator alerts, or experiment terminations.

Founding President, IEEE EMBS Student Chapter — Mar. 2025 – May 2026
University of Florida — *Gainesville, Fl*
- Managed a team of 18 officers to plan 10+ academic, technical, and professional workshops, speaker events, and state-wide competitions for 150+ students to support awareness and professional development in interdisciplinary fields.

Research and Development Intern — Jun. 2025 – Aug. 2025
Pfizer — *Groton, CT*
- Coordinated DMAIC objectives by eliciting data from employee interviews to create 2 process maps, leading to the creation of 3 subteams for improvement implementation of 10+ areas of the API supply planning chain.

RODI Data Science Co-op Intern — Jan. 2025 – Jun. 2025
Johnson & Johnson — *Remote*
- Completed 1st PoC run of operational submissions simulation using discrete event modeling platform to greenlight project.

Undergraduate Research Assistant — Jan. 2023 – Dec. 2024
SMILE Lab — *Gainesville, FL*
- Trained the SynthSeg deep learning tool on 150+ full head MRI scans from the clinical study on cognitive training augmentation to determine robustness on scans that contained data on 11 whole-head tissue classes.
- Processed 3 MRI datasets with 9 transcranial electrode configurations in a physics-based electric field simulation model to be utilized in a denoising diffusion model for creating a brain digital twin to perform precision dosing of tDCS.

Publications & Posters

Veronica Ramos Rodriguez, et al. "Comparison of GRACE and SynthSeg Deep Learning Models for Whole-Head Brain Segmentation," BMES, Baltimore, Maryland, Oct. 2024.

Eden Dalcourt, **Veronica Ramos Rodriguez**, et al. "Overcoming Obstacles for the Distribution of Advanced Computer Simulation Devices in Medicine," *International Journal of Student Project Reporting.* DOI: 10.1504/IJSPR.2023.10056644.

Projects

Unsupervised Machine Learning Kidney Cell Clustering | *Python* — Mar. 2024 – May 2024
- Utilized OpenCV, Scikit-learn, and UMAP for generalized image segmentation and unsupervised classification of 180,000+ nuclei for kidney and lung DAPI and H&E whole-slide images.

Skills

Technical: Linux, Git, KiCAD, LTspice, Cadence Virtuoso, Android Studio, Arena Simulation, Lean Six Sigma (Yellow), GPUs
Programming: Python, MATLAB, Rust, C++, C#, Java, Shell
Libraries: pandas, NumPy, scikit-learn, TensorFlow, Keras, PyTorch, OpenCV, OpenGL ES, NEURON, LFPy